Comment in Response to Request for Information: Development of an Artificial Intelligence (AI) Action Plan

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Introduction

We are submitting this comment in response to the Office of Science and Technology Policy's (OSTP) Request for Information on the Development of an Artificial Intelligence (AI) Action Plan. We are both affiliated with the AI Guide Dog (AIGD) research project at **Carnegie Mellon University**, where we focus on developing AI-powered assistive technologies that address accessibility challenges faced by individuals with disabilities. The views expressed in this comment are our own and do not represent the positions or policies of any institutions with which we are affiliated.

Our research, the AI Guide Dog project, centers on an AI-driven smartphone navigation assistant designed to empower blind individuals with real-time, reliable navigation support. Developed through a collaboration between Carnegie Mellon University and industry researchers, AIGD enables users to confidently and independently navigate daily environments—without the need for bulky or intrusive hardware—lowering both financial and technological barriers to independent mobility.

We strongly believe that the U.S. AI Action Plan should include strategic support and incentives for AI research and development initiatives aimed at improving citizens' well-being and creating inclusive communities through accessible technologies.

Background

We appreciate the Administration's recognition of AI as a tool to promote human flourishing and economic competitiveness, as reflected in Executive Order 14179. In line with these objectives, we believe AI-powered assistive technologies represent an important area where AI can directly enhance Americans' quality of life, particularly for individuals with disabilities.

AI-driven accessibility solutions can:

- Improve **health and well-being** by fostering independence and confidence in daily activities.
- Reduce **inequality** by addressing mobility barriers faced by visually impaired and mobility-impaired individuals.

- Support **sustainable and inclusive communities** by providing valuable data for accessible urban infrastructure planning.

Beyond individual empowerment, accessible AI technologies offer significant **economic and societal benefits**. In 2018, potential productivity losses resulting from the exclusion of blind individuals from full participation in society were estimated at **$410 billion annually**, representing **0.3% of the world's GDP**, as noted by Dr. Chieko Asakawa, IBM [2]. By enabling greater participation through AI and robotics, we can unlock economic productivity and foster a more inclusive, prosperous future.

Despite significant advancements in AI capabilities, the widespread adoption of AI assistive technologies remains limited due to critical barriers—including prohibitive costs, social stigmatization, and urban infrastructure constraints.

Analysis

Through our work, AI Guide Dog: Smartphone-Based Navigation Assistant for the Blind [1], we identified several systemic barriers that hinder the broader adoption of AI-powered assistive technologies:

- **Intrusiveness**: Many systems require specialized, bulky, or socially intrusive hardware.
- **Steep learning curves**: Users often require extensive training to adapt to complex systems.
- **Social acceptance**: Navigating public spaces with conspicuous devices can be stigmatizing.
- **Infrastructure limitations**: Poorly designed urban environments make it difficult for assistive technologies to function effectively.
- **Explainability and reliability**: AI systems must provide transparent, predictable, and trustworthy guidance to build user confidence.
- **Lack of standardized safety regulations and testing protocols** governing the development and deployment of AI-powered mobility aids. Without unified standards, it becomes difficult to certify these systems as safe and effective for public use.

Our research has also produced **valuable data-driven insights into** blind mobility patterns, and pedestrian behavior, with important implications for **urban planning**, **infrastructure development**, and **public safety**. This data can:

- Identify challenging navigation areas requiring infrastructure improvements, such as poorly designed sidewalks, crosswalks, or intersections.
- Optimize pedestrian pathways to enhance overall accessibility.
- Enhance public transit integration through real-time accessibility information.

Collaboration with **policy makers, healthcare professionals,** and **city planners** can further extend the impact of these insights by:

- Developing safety guidelines for shared spaces.
- Informing emergency evacuation protocols for blind individuals.
- Supporting the creation of mobility training and therapeutic programs to promote independent navigation.

Despite the availability of research and data like ours, current policy frameworks lack sufficient support for AI accessibility initiatives. As a result, promising innovations such as AIGD face significant hurdles in progressing from research to widespread deployment.

Recommendations

We respectfully submit the following **concrete policy actions** for consideration in the AI Action Plan:

1. **Establish a Federal AI for Accessibility Initiative**
   Provide dedicated funding and incentives for AI research focused on assistive technologies that address mobility and accessibility challenges for people with disabilities.

2. **Develop National Safety and Performance Standards for AI Assistive Technologies**
   Convene a multi-stakeholder working group—including NIST, ANSI, AI researchers, disability advocates, and urban planners—to develop standardized testing protocols, safety benchmarks, and certification processes for AI-powered mobility systems.

3. **Create a Public-Private Urban Accessibility Data Consortium**
   Support data-sharing initiatives that leverage insights from AI-assisted navigation systems to identify infrastructure gaps and inform inclusive urban planning, transportation design, and emergency preparedness strategies.

4. **Offer Tax Incentives and Grants for Accessible AI Product Development**
   Provide financial support for companies and research groups developing affordable, non-intrusive AI mobility solutions and subsidize the cost of these devices for end users.

5. **Invest in AI Accessibility Workforce Development**
   Fund education and training programs that empower people with disabilities to engage with AI technologies as users, testers, and developers, promoting greater

inclusion in the AI workforce.

Conclusion

AI-powered assistive technologies hold immense potential to improve the lives of Americans with disabilities, foster inclusive communities, and deliver broader societal and economic benefits. However, realizing this potential requires **targeted government support**, **robust safety standards**, and **multi-stakeholder collaboration**. We urge the OSTP and NITRD NCO to incorporate these recommendations into the AI Action Plan to ensure AI innovation is inclusive, equitable, and beneficial to all Americans.

Sincerely,
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References

[1] Jadhav, A, et al. AI Guide Dog: Egocentric Path Prediction on Smartphone. Paper accepted in AAAI Spring Symposium, 2025. https://doi.org/10.48550/arXiv.2501.07957

[2] Chieko A, IBM. Empowering the Visually Impaired with AI Technology. Retrieved from: https://aiforgood.itu.int/empowering-the-visually-impaired-with-ai-technology/